

1(a)	mass / volume	B1
1(b)(i)	(vernier/digital) calipers	B1
1(b)(ii)	percentage uncertainty = $(0.0004 / 0.0420) \times 100$ = 1%	A1
1(c)(i)	$\text{kg m}^{-3} = \text{kg} \times \text{m}^n / \text{m}$ or $\text{kg m}^{-3} = \text{kg} \times \text{m}^n \times \text{m}^{-1}$	M1
	$-3 = n - 1$ and (so) $n = -2$	A1
1(c)(ii)	$(\Delta\rho/\rho) = (\Delta M/M) + 2(\Delta r/r) + (\Delta L/L)$	C1
	percentage uncertainty = $[(0.001 / 1.072) + 2 \times (0.0004 / 0.0420) + (0.0001 / 0.1242)] (\times 100)$	C1
	= 0.09% + 2 × 0.95% + 0.08% = 2%	A1
1(c)(iii)	$\rho = (1.072 \times 0.0420^{-2}) / (2.094 \times 0.1242)$ = 2337 (kg m ⁻³)	C1
	$\Delta\rho = 0.021 \times 2337$ = 49 (kg m ⁻³)	C1
	$\rho = (2340 \pm 50) \text{ kg m}^{-3}$	A1